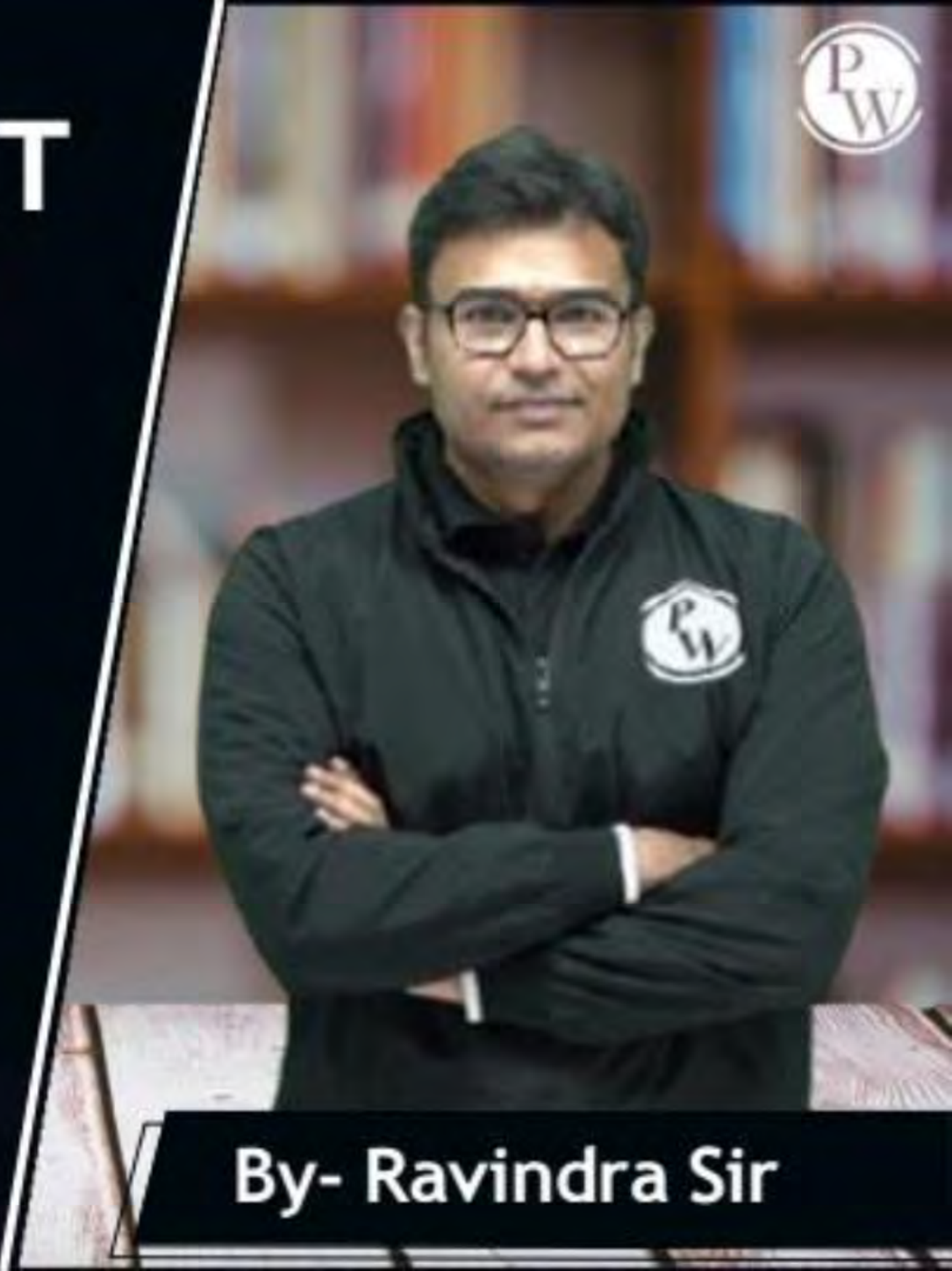
Computer Science & IT

COMPUTER NETWORK
(CN)

PYQ - CN

Lecture No. 02



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Recap of Previous Lecture



- Topic
- Topic
- Topic

PYQ

Topics to be Covered



- Topic
- Topic
- Topic

PYQ

In a sliding window ARQ Scheme, the transmitter's window size is N and Receivers window size is m. The min no of distinct sequence numbers required to ensure correct operation of the ARQ scheme

a) $\min(m, N)$ b) $\max(m, n)$ c) $m+N$ d) $m*N$





N+m Seq numbers

A 20 kbps link has a propagation delay of 400 ms. The transmitter employs the "go back N" ARQ with n set to 10. Assuming that each frame is 100 bytes long what is the max data rate possible?

a) 5 kbps b) 10 kbps c) 15 kbps d) 20 kbps

$a = \frac{T_p}{T_t}$ $T_p = 400\,ms$ $T_t = \frac{L}{B} = \frac{800\,bits}{20 \times 10^3\,bps} = 40\,msec$

$a = \frac{400}{40} = 10$

$Th = \eta * BW = \frac{N}{1+2a} * BW = \frac{10}{1+20} \times 20\,kbps$

$= \frac{10}{20} \times 20\,kbps$

$= 10\,kbps$

The maximum window size for data transmission using the Selective reject protocol with n bit frame sequence numbers is

a) 2^n b) 2^{n-1} c) $2^n - 1$ d) $2^n - 2$

2^n seq number

$W_S = W_R$

$2^n \quad \frac{2^n}{2} \quad 2^{n-1}$

Consider the following message $M = 1010001101$. The Cyclic Redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is

a) 01110 b) 01011 c) 10101 d) 10110

10100011010

$x^5 + x^4 + x^2 + 1$

$1 \cdot x^5 + 1 \cdot x^4 + 0 \cdot x^3 + 1 \cdot x^2 + 0 \cdot x^1 + 1 x^0$

```
110101 ) 10100011010 00000
         110101
         ------
         0111011010 00000
           110101
           ------
           0011101010 00000
             110101
             ------
             0011111100 00000
               110101
               ------
               0010110000 0
                 110101
                 ------
                 0110010 00
                   110101
                   ------
                   0011 10
```

A channel has a bit rate of 4 kbps and one way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of ACK is negligible. To get an efficiency of 50 % the minimum frame size should be

a) 80 bytes b) 80 bits c) 160 B d) 160 b

$$\eta = \frac{1}{1+2a} \geq 1/2 \Rightarrow 2 \geq 1+2a \Rightarrow 2a \leq 1 \Rightarrow 2 * \frac{TP}{Tt} \leq 1$$

$$2 * \frac{20 \times 10^{-3}}{L} * 4 \times 10^{3} \leq 1$$

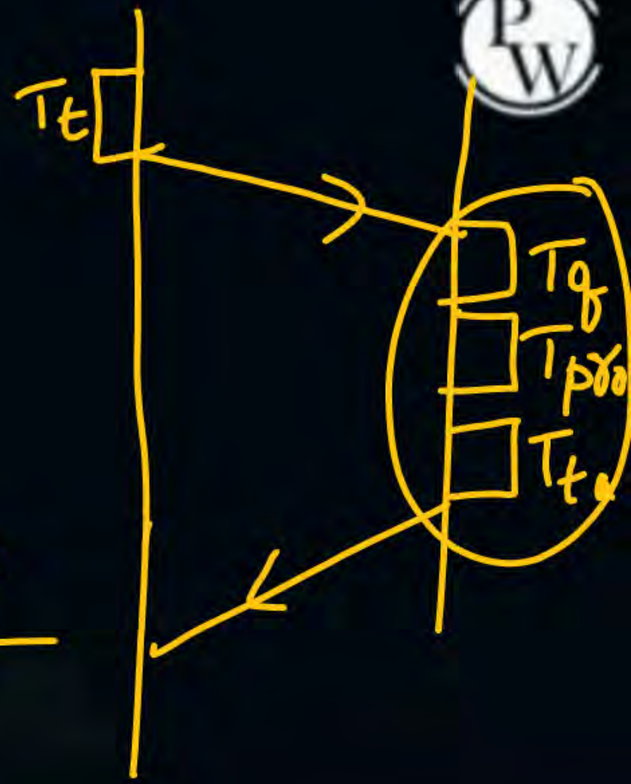
$$\Rightarrow 160 \leq L \Rightarrow L \geq 160 \text{ bits}.$$



$$T_t \geq 2 * T_p$$

$$\frac{L}{B} \geq 2 * T_p$$

$$L \geq 2 * T_p \times B$$



$$\eta = \frac{T_t}{T_t + T_q + T_{proc} + T_{transact} + 2 * T_p}$$

→ on a wireless link, the probability of packet error is 0.2. A stop and wait protocol is used to transfer data. The channel condition is assumed to be independent from transmission to transmission. The average no of transmission attempts required to transfer 100 packets

a) 100 b) 125 c) 150 d) 200

$$n + np + np^2 + \cdots$$

$$n(1 + p + p^2 + p^3 + \cdots)$$

$$\frac{n}{1-p} = \frac{100}{0.8} = 125$$

Station A uses 32 byte packets to transmit to station B using sliding window protocol. The round trip delay b/w A and B is 80 msec, and BW on the path b/w A and B is 128 Kbps. what Should be the optimal window size 'A' should use?

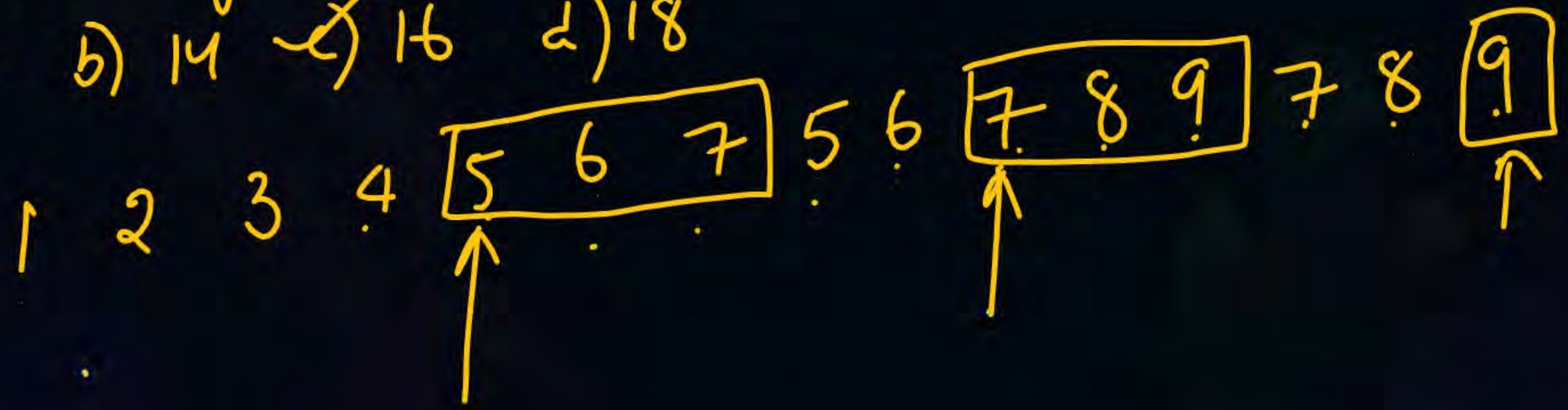
a) 20 b) 40 c) 160 d) 320

$$W_S = 1 + 2a = 1 + \frac{2 * T_p}{T_t} = 1 + \frac{80 \times 10^{-3} \text{ sec}}{\frac{32 \times 8}{128 \times 10^3}}$$

$$= 40$$

Station A needs to send a message consisting of 9 packets to station B using SWP (window size = 3) and GBN strategy. If every 5th packet that A transmits gets lost, then what is the number of packets that A will transmit for sending the message to B?

a) 12 b) 14 c) 16 d) 18



The message 11001001 is to be transmitted using CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is $1x^3+0\cdot x^2+0\cdot x^1+1x^0$

a) 11001001000

b) 11001001011

c) 11001010

d) 11001001011

1001) 11001001000

1001

01011001000

1001

00100001000

1001

00011060

1001

0|010

100 1

0011

1001

1001

0|0 1 1

In the following pairs of OSI protocol layer/sublayer and its functionality, the incorrect pair is

✓a) N/W layer and Routing

Xb) DLL and bit Synchronization

✓c) Transport layer and End to end communication

✓d) Medium Access control Sublayer and channel sharing

A 1mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/sec. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window protocol? Assume that the Ack packets are negligible in size and there are no errors during communication

a) 120B b) 60B c) 240B d) 90B

$d = 36,504 \times 10^3$ $\quad v = 3 \times 10^8$ m/sec

$$\eta = \frac{N}{1+2a} \Rightarrow \frac{1}{4} = \frac{127}{1+2 \times \frac{36,504 \times 10^3}{3 \times 10^8} \times \frac{10^6}{L}}$$

$$\Rightarrow L = 120B$$

THANK - YOU